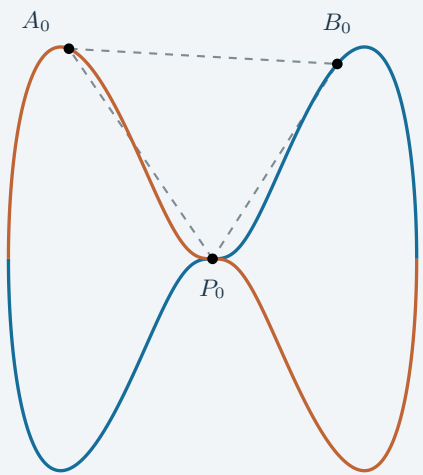
Two quadratic transformations

$C_0 : \ (YZ - X^2)^2 - Y^3 Z = 0$

Schematic global views: all singularities shown on each connected curve

C_0: triple contact A_5

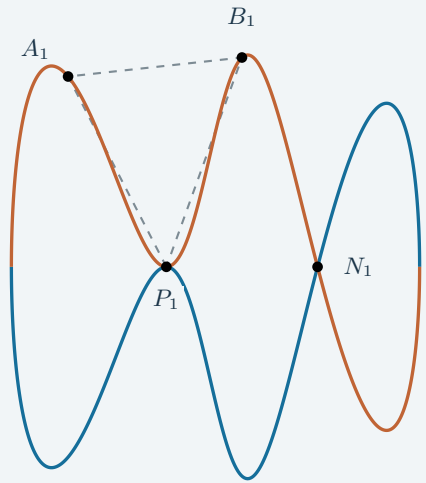


φ_1 centered at $\{P_0, A_0, B_0\}$

$\mathrm{Sing}(C_0) = \{P_0\}$

φ_1

C_1: tacnode + one node



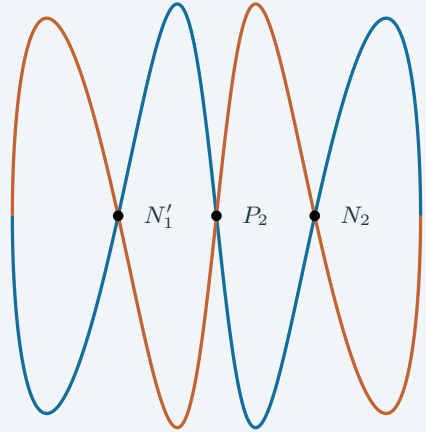
$N_1 = q_1(\overline{A_0 B_0})$

q_2 centered at $\{P_1, A_1, B_1\}$

$\mathrm{Sing}(C_1) = \{P_1, N_1\}$

φ_2

C_2: three ordinary nodes



$N_1' = \varphi_2(N_1)$

$N_2 = \varphi_2(\overline{A_1 B_1})$

$\mathrm{Sing}(C_2) = \{P_2, N_1', N_2\}$